

VUNISEA, Fiji

WMO: 916930

Lat: 19.050S

Lon: 178.167E

Elev: 108

StdP: 14.64

Time Zone: 12.00 (PCF)

Period: 86-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	64.2	66.1	55.3	65.4	70.4	57.1	69.7	71.1	24.8	74.4	22.8	74.4	1.0	120	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.1	87.6	79.4	86.8	79.0	85.9	78.6	80.9	85.8	80.1	85.0	79.5	84.3	7.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.4	152.9	84.4	78.8	150.1	83.8	78.1	146.5	83.1	44.7	86.4	44.0	85.4	43.2	84.7	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	20.5	18.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.2	80.1	80.9	80.6	79.4	76.6	75.0	73.5	73.4	74.4	75.4	77.6	79.4
	DBStd	3.43	1.63	1.63	1.71	1.90	2.02	2.28	2.13	2.45	2.90	2.25	2.08	1.92
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9917	933	866	948	883	823	750	729	727	731	786	829	912
	CDD65	4442	468	446	483	433	358	300	264	262	281	321	379	447
	CDH74	31193	4396	4476	4664	3708	2020	1237	763	809	1187	1477	2603	3853
	CDH80	5528	959	1198	1132	673	167	72	32	34	90	108	387	676
(14) Wind	WSAvg	7.6	7.3	6.3	6.4	7.6	8.0	7.9	7.1	7.0	8.2	8.7	7.5	9.1
(15) Precipitation	PrecAvg	83.2	11.3	9.6	10.6	9.1	6.3	4.6	3.7	3.7	4.5	5.5	6.3	8.2
	PrecMax	116.2	26.0	16.7	16.4	16.1	14.2	11.4	6.4	9.7	8.6	10.4	16.0	15.5
	PrecMin	56.1	2.2	2.5	4.8	2.7	1.8	1.5	0.9	0.9	0.7	1.4	2.0	2.8
	PrecStd	15.2	5.6	3.6	3.2	3.9	3.2	2.5	1.5	2.1	2.2	2.4	3.1	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	88.8	88.6	88.0	84.7	83.7	82.7	82.8	85.0	84.5	86.7	87.8
		MCWB	79.4	80.0	80.0	79.7	77.1	76.5	75.1	74.9	76.6	76.3	77.8	79.2
	2%	DB	86.6	87.4	87.2	86.3	83.1	82.0	80.6	80.9	82.4	82.5	85.0	86.3
		MCWB	78.8	79.6	79.4	78.8	76.1	75.9	73.9	74.1	75.6	75.2	77.1	78.8
	5%	DB	85.5	86.5	86.2	85.0	81.9	80.5	78.9	79.5	80.8	81.1	83.6	84.7
		MCWB	78.2	79.1	78.9	78.0	75.2	74.9	72.4	73.2	74.5	74.1	76.4	77.8
	10%	DB	84.5	85.5	85.1	83.7	80.7	79.1	77.6	78.0	79.1	79.8	82.2	83.4
		MCWB	77.6	78.6	78.4	77.2	74.5	73.8	71.5	72.2	73.2	73.1	75.6	76.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	82.0	81.9	81.2	79.0	78.0	77.3	76.5	77.8	78.5	79.7	81.0
		MCDB	85.9	86.5	86.4	86.1	82.8	81.8	80.2	81.2	82.7	82.3	84.6	85.7
	2%	WB	80.0	80.8	80.7	79.9	77.8	76.7	75.3	75.3	76.5	76.9	78.3	79.7
		MCDB	84.9	85.9	85.5	84.8	81.6	80.8	79.0	79.4	81.2	80.9	83.1	84.5
	5%	WB	79.1	80.1	79.9	79.0	76.8	75.8	74.2	74.3	75.4	75.6	77.4	78.8
		MCDB	84.0	85.1	84.6	83.6	80.6	79.5	77.8	78.3	79.7	79.4	82.0	83.4
	10%	WB	78.4	79.3	79.1	78.2	76.0	74.8	73.2	73.3	74.4	74.5	76.5	78.0
		MCDB	83.2	84.2	83.8	82.5	79.5	78.2	76.6	77.2	78.4	78.4	80.9	82.3
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.1	8.8	7.9	7.8	7.4	8.4	8.5	8.2	8.5	8.5	7.7
		MCDBR	10.2	10.6	10.7	10.2	9.8	9.7	10.6	10.5	9.9	10.3	10.7	9.5
	5% WB	MCWBR	5.1	5.1	5.2	5.1	5.2	5.4	5.5	5.5	5.6	5.5	5.4	5.0
		MCWBR	9.5	9.9	9.8	9.3	8.4	8.9	9.2	9.1	9.3	9.1	9.8	8.6
(40) Clear-Sky Solar Irradiance	taub		0.381	0.383	0.379	0.375	0.369	0.369	0.359	0.359	0.373	0.378	0.376	0.379
		taud	2.639	2.638	2.663	2.662	2.635	2.623	2.634	2.643	2.590	2.591	2.623	2.632
	Ebn at Noon		305	302	296	285	275	269	275	286	293	299	305	306
		Edh at Noon	32	31	30	28	27	26	26	28	31	33	32	32
(44) All-Sky Solar Radiation	RadAvg		2043	1953	1754	1467	1281	1128	1204	1370	1628	1807	1948	2021
	RadStd		179	130	107	110	87	86	66	128	150	135	182	107

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.32	N/A	N/A	N/A	N/A	+0.36	+0.43	N/A	N/A	+121	+119

Nomenclature: See separate page